LaTeX Author Guidelines for the ACS FUB Proceedings Manuscripts

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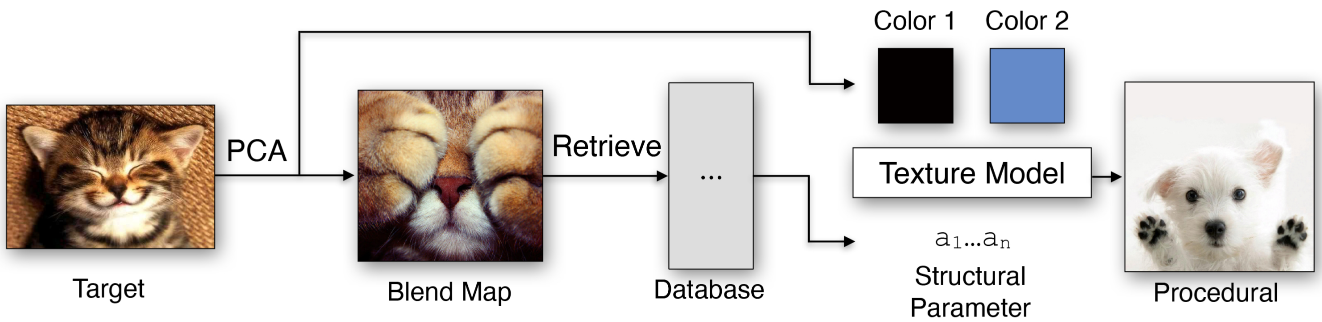

Figure 1: *Teaser Image. Explain the image.*

Abstract

*The abstract should be between 100-150 words and should be a succinct and stand alone description. An abstract is neither a summary nor an outline of the paper. For example, you can follow these four sentences [JBB*93]: State the problem, Why is it a problem and why is it interesting? Your solution and achievements. What follows from your solution?*

1. Introduction

Generative AI for synthetic image generation has become increasingly popular. While those models are supporting multiple kinds of commercial and private workflows, copyright as well as ethical concerns regarding the usage of GenAI images are becoming progressively urgent. Political mis- and disinformation is more than ever supported by the usage of GenAI imagery. Since there are no signs of GenAI usage decreasing in the future, rather the opposite, it becomes necessary to be able to distinguish GenAI images from images which were not produced by GenAI.

To address the socio-political problems derived by the usage of GenAI imagery described above, multiple approaches to distinguish between conventional images (non-GenAI images) and synthetic (GenAI images) are discussed. Adding an invisible watermark to synthetic images, is one of the most recent, actively researched methods to help enable this distinction.

For the generation of those images, Generative Image Diffusion Models have become, next to GANs, the most common and therefore the industry standard. Their output makes up the largest amount of GenAI images produced. Their generation method works with progressively diffusing image samples with Gaussian noise and then reverse that process to generate output images. Large and popular Image Generation models like StableDiffusion, Midjourney as well as Dall-E3 and Imagen are diffusion models.

Invisible watermarking methods of diffusion model images usually alter the frequency domain of the images, during or after the generation process. Therefore this paper concentrates on invisible watermarking methods for images of generative diffusion models that work on the image's frequency domain.

To provide practical understanding of how image watermarking methods can be evaluated, the paper will look at three different methods on how to invisibly watermark GenAI images by altering the frequency domain. The first method is developed by Chen et al. and described in their paper Image Watermarking of Generative Diffusion Models (2025). They developed a watermarking method which implements a reconstructable image watermark during the training process of the model. The second method by Wei et al from the paper DiffKGW: Stealthy and Robust Diffusion Model Watermarking (2026) take on the watermarking method of LLMs by subtly biasing which noise values are sampled at the start of image generation. The third method ... (t.b.d.). All three approaches will be compared by an evaluation set consisting of four aspects:

- Robustness: The preservation of the watermark against attacks.
- Capacity: The amount of information that can be hidden with the watermark.
- Imperceptibility: The level of altering of the output image quality, compared to clean images.
- Implementation effort: The resources needed to integrate the method into a model.

This comparison is intended to give an overview on how different watermarking strategies for diffusion models working with the frequency domain are discussed, how they differ and what their individual advantages are. Eventually, this paper aims to discuss potential technical obstacles hindering practical implementation of those methods into real-world scenarios, to support ongoing research on invisible watermarking of diffusion model's output.

2. Related Work

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3. Main Content - Name and Structure the Main Content According to your Paper

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3.1. Method

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3.2. Results

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3.3. Discussion

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4. Formatting your paper

4.1. Bullet Points

The following is an example for a bullet points list. The contributions of this paper are

- my greate archivement number one,
- a second awsome insight,
- and last but not least the third contribution.

4.2. Figures

All graphics should be centered.

If your paper includes images, it is important that they are of sufficient resolution to be faithfully reproduced.

Figure 2: *Here is a sample figure. All figures must have sensible captions and must be referenced in the text.*

Instructions from the original conference template (for the ACS FUB it is not that relevant):

To determine the optimum size (width and height) of an image, measure the image's size as it appears in your document (in millimeters), and then multiply those two values by 12. The resulting values are the optimum x and y resolution, in pixels, of the image. Image quality will suffer if these guidelines are not followed.

Example 1: An image measures 50 mm by 75 mm when placed in a document. This image should have a resolution of no less than 600 pixels by 900 pixels in order to be reproduced faithfully.

This is an examplary reference to Figure 3, which shows the incredible test image from the Eurographics conference.

4.3. Tables

A simple table might look like the following. However there are several ways to create and layout tables - it is up to you to style a table up to your liking. This is a reference to Table 1.

4.4. Formulas

In LaTeX it is fairly easy to add mathematical symbols and formulas. See the following as example [wik19]:

Col1	Col2	Col2	Col3
1	6	87837	787
2	7	78	5415
3	545	778	7507
4	545	18744	7560
5	88	788	6344

Table 1: *Here is a sample table. All tables must have sensible captions and must be referenced in the text.*

$$a = bq + r \tag{1}$$

which is true if a and b are integers with $b \neq c$.

$$L' = L\sqrt{1 - \frac{v^2}{c^2}} \tag{2}$$

4.5. Footnotes

Please do not use footnotes at all.

4.6. References

References are layouted automatically. Add the needed entry to the bibtex file and use the citation command to reference the entry.

If you don't want to mention the author's name just add the citation to the end of the sentence [Per85]. You can also cite more than one references at the same time [GSH18, Per85, SLKD15].

Sometimes you want to refer to the authors name. With this template you need to write out the names manually. For more than two authors you only use the name of the first author and abbrivate the remaining names with "et al."

An example for more than two authors:

Raul et al. [RRDR09] thoroughly review the different approaches for segmentation algorithms. Our problem requires the segmentation of color images, summarized by Cheng et al. [CJSW01].

An example for two authors:

Also aiming for a fragmented aesthetic appeal by transforming original image data "piecewise", Collomosse and Hall [CH03] render from salient image features a cubist version of an input. Similarly, Lai and Rosin [LR13] loosely aim for a cubist appearance by segmenting an input and by simplifying the segments into constant colors, while maintaining global structures.

An example for one author:

Artistically, we are inspired by the art of Wehrli [Weh03], who manually creates image decompositions and rearrangements.

5. Conclusion

Please direct any questions to the production editor in charge of these proceedings.

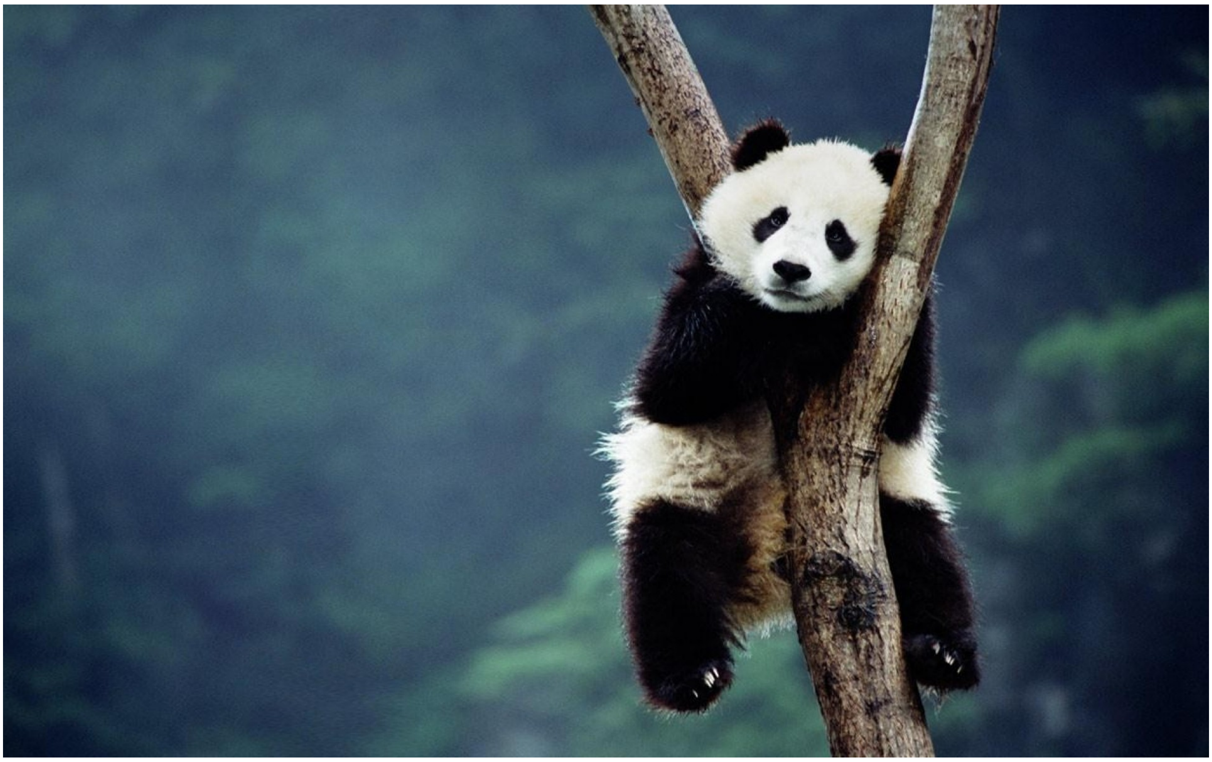

Figure 3: *Two-column width figure*

References

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